DEPI Graduation Project Submission

EduHub Education Platform

User Case Descriptions &Functional Requirements

Submitted By

1. Aya Abdelmoniem Mohamed Hamed	2. Ibrahim Mohamed Ibrahim Abdo
3. Reem Ashraf Mohammed Ahmed	4. Ahmed Ramadan Ali Mohamed
5. Warda Mahmoud Khalaf Ahmed	6. Mohamed Nashaat Elsayed Elashrey
7. Osama Sharkawy Abady Mohamed	

Supervised By

Dr Ahmed Samir

1. Project Introduction

1.1 Project Overview

EduHub is a comprehensive, modern web-based Online Learning Platform engineered to connect students and instructors through interactive online courses, quizzes, final examinations, and live learning sessions. The system provides robust capabilities for course creation, dynamic content delivery, progress tracking, automated grading, certificate issuance, financial earnings processing, and administrative moderation.

1.2 Core System Capabilities & Scope

The platform encompasses the following core functional areas:

- **User Authentication & Security:** Role-based access control (Student, Instructor, Admin), secure registration, login, profile management, and OAuth integration.
- **Taxonomy & Preferences:** Structured categories and subcategories with customizable student interest preferences.
- **Course Management & Delivery:** Multi-section and multi-lesson course hierarchy (supporting free previews, public catalogs, and structured syllabi).
- **Quality Assurance & Workflow:** Administrative course review queue, approval/rejection workflows with detailed reason tracking, and active status moderation.
- **Financial & Payment Infrastructure:** Payment integration via Stripe Hosted Checkout, transactional commission splitting, financial audit logs, and automated refund processing with access revocation.
- **Assessment & Certification Engine:** Section quizzes with attempt limits (default 3 attempts), mandatory single-attempt final examinations, automated grading, score auditing, and unique verifiable PDF certificates.
- **Live Sessions:** Integration with external meeting services (Zoom, Google Meet, Microsoft Teams) for real-time virtual classrooms.

2. Functional Requirements

This section lists all 135 Functional Requirements (REQ-01 to REQ-135) governing the EduHub platform.

Module 1: User Registration, Authentication & Account Management

ID	Requirement	Description
REQ-01	User Registration	The system shall allow visitors to create an account by providing Full Name, Email, Password, and Role (Student/Instructor).
REQ-02	Email Verification	The system shall send a verification token email upon registration and require confirmation before full account activation.
REQ-03	User Authentication	The system shall authenticate users using registered credentials and return secure JWT access and refresh tokens.
REQ-04	Password Reset Request	The system shall allow users to request a password reset link via their registered email address.
REQ-05	Password Reset Completion	The system shall validate password reset tokens and update the password hash in the database.
REQ-06	User Logout	The system shall invalidate active refresh tokens and clear client-side session states upon logout.
REQ-07	Token Refresh	The system shall issue new JWT access tokens using valid refresh tokens without requiring credential re-entry.
REQ-08	OAuth Sign-In	The system shall support single sign-on authentication via Google OAuth2 integration.
REQ-09	Role Assignment	The system shall enforce role mapping (Student, Instructor, Admin) stored in the database.
REQ-10	RBAC Authorization	The system shall restrict API endpoint access based on assigned user roles.
REQ-11	Instructor Application	The system shall allow Students to submit an application with credentials and bio to upgrade to Instructor role.
REQ-12	Application Review	The system shall provide Admins with an interface to approve or reject pending Instructor applications.
REQ-13	Account Deactivation	The system shall allow Admins to suspend user accounts, blocking login and access.
REQ-14	Account Reactivation	The system shall allow Admins to restore previously suspended user accounts.

Module 2: User Profile Management

ID	Requirement	Description
REQ-15	View Profile	The system shall allow logged-in users to view their complete account profile details.
REQ-16	Update Profile	The system shall allow users to update display name, headline, biography, and social media links.
REQ-17	Avatar Upload	The system shall allow users to upload and crop custom profile avatar images (JPEG/PNG, max 2MB).
REQ-18	Password Change	The system shall allow authenticated users to change their password by confirming their current password.
REQ-19	Public Instructor Page	The system shall render public instructor profiles displaying bio, metrics, and published courses.
REQ-20	Notification Preferences	The system shall allow users to toggle email notification settings for announcements and course updates.

Module 3: Student Preferences and Category Mapping

ID	Requirement	Description
REQ-21	Select Preferences	The system shall allow Students to select multiple preferred course categories during or after onboarding.
REQ-22	Update Preferences	The system shall allow Students to modify saved category preferences anytime from settings.
REQ-23	Personalized Recommendations	The system shall generate course recommendations on the student dashboard matching saved preferences.
REQ-24	Onboarding Preference Prompt	The system shall prompt newly registered Students to select category preferences upon first login.
REQ-25	Preference-Based Alerts	The system shall notify Students when new courses are published in their preferred categories.

Module 4: Course Discovery and Browsing

ID	Requirement	Description
REQ-26	Browse Catalog	The system shall display a public catalog of all published courses with title, thumbnail, rating, and price.
REQ-27	Filter by Category	The system shall allow users to filter catalog courses by primary Category and child Subcategory.
REQ-28	Filter by Price	The system shall allow filtering by course pricing type (Free vs. Paid).
REQ-29	Filter by Rating	The system shall allow filtering by minimum star rating (e.g., 4.0+, 4.5+).
REQ-30	Search Keywords	The system shall provide full-text keyword search across course titles, subtitles, and descriptions.
REQ-31	Catalog Sorting	The system shall allow sorting catalog results by popularity, newest release, rating, and price.
REQ-32	Featured Courses	The system shall render an Admin-curated list of featured courses on the home landing page.
REQ-33	Catalog Pagination	The system shall paginate catalog query results to optimize load performance.

Module 5: Course Overview and Preview Details

ID	Requirement	Description
REQ-34	Course Overview Page	The system shall display comprehensive course details including syllabus tree, outcomes, and requirements.
REQ-35	Syllabus Inspection	The system shall expose section titles and lesson titles without granting video playback to non-enrolled users.
REQ-36	Free Sample Preview	The system shall allow any visitor to stream video lessons flagged as IsFreePreview=True.
REQ-37	Instructor Overview Info	The system shall render instructor profile snippets on the course detail landing page.
REQ-38	Aggregate Reviews Display	The system shall display average star ratings, rating breakdowns, and student text reviews on the course page.
REQ-39	Enrolled Student Count	The system shall display the total number of enrolled students on the course landing card.
REQ-40	Course Prerequisites	The system shall list requirements and targeted target audiences specified by the instructor.

Module 6: Course Creation and Authoring

ID	Requirement	Description
REQ-41	Create Draft Course	The system shall allow Instructors to create new course entities initialized with Status='Draft'.
REQ-42	Update Course Metadata	The system shall allow Instructors to update draft course title, subtitle, description, language, and level.
REQ-43	Configure Course Pricing	The system shall allow Instructors to define course price or mark the course as Free.
REQ-44	Upload Course Cover	The system shall allow Instructors to upload custom course thumbnail image banners.
REQ-45	Assign Taxonomy	The system shall enforce selecting exactly one parent Category and one Subcategory per course.
REQ-46	Submit for Approval	The system shall allow Instructors to submit draft courses for Admin review when minimum syllabus rules are met.
REQ-47	Delete Draft Course	The system shall allow Instructors to delete owned course drafts that have zero enrollments.
REQ-48	Archive Course	The system shall allow Instructors or Admins to archive published courses, hiding them from new enrollments.
REQ-49	Instructor Dashboard	The system shall provide Instructors with a dashboard listing all authored courses and their statuses.
REQ-50	Resubmit Rejected Course	The system shall allow Instructors to modify and resubmit courses previously rejected by Admins.

Module 7: Syllabus Structure – Sections & Lessons

ID	Requirement	Description
REQ-51	Create Section	The system shall allow Instructors to add named curriculum sections to a course draft.
REQ-52	Edit Section	The system shall allow Instructors to update section titles and descriptions.
REQ-53	Reorder Sections	The system shall allow Instructors to reorder section positions using drag-and-drop sequencing.
REQ-54	Delete Section	The system shall allow Instructors to delete empty sections or cascade delete associated draft lessons.
REQ-55	Create Lesson	The system shall allow Instructors to create lessons within a section specifying title and content type.
REQ-56	Upload Lesson Video	The system shall process and store uploaded lesson video media files and calculate video duration.
REQ-57	Toggle Free Preview	The system shall allow Instructors to mark specific lessons as free preview items.
REQ-58	Reorder Lessons	The system shall allow Instructors to reorder lessons within a section.

Module 8: Category & Subcategory Administration

ID	Requirement	Description
REQ-59	Create Category	The system shall allow Admins to create new top-level categories with name, slug, and icon.
REQ-60	Edit Category	The system shall allow Admins to edit existing category attributes.
REQ-61	Delete Category	The system shall allow Admins to delete categories that contain zero subcategories or courses.
REQ-62	Create Subcategory	The system shall allow Admins to create child subcategories linked to a parent category.
REQ-63	Edit Subcategory	The system shall allow Admins to edit subcategory names and slugs.
REQ-64	Delete Subcategory	The system shall allow Admins to delete subcategories that contain zero assigned courses.

Module 9: Course Approval & Moderation Workflow

ID	Requirement	Description
REQ-65	Approval Queue	The system shall provide Admins with an inspection queue listing all courses with Status='PendingApproval'.
REQ-66	Inspect Submission	The system shall allow Admins to inspect complete video content, syllabus, and metadata of pending courses.

ID	Requirement	Description
REQ-67	Approve Course	The system shall allow Admins to approve pending courses, transitioning status to Published.
REQ-68	Reject Course	The system shall allow Admins to reject pending courses, requiring text entry into RejectionReason.
REQ-69	Unpublish Course	The system shall allow Admins to unpublish an active course for terms violations.
REQ-70	Clear Rejection Log	The system shall automatically clear previous RejectionReason logs upon successful instructor resubmission.
REQ-71	Status Email Alerts	The system shall dispatch email notifications to instructors whenever course approval status changes.

Module 10: Student Enrollment Processing

ID	Requirement	Description
REQ-72	Enroll Free Course	The system shall immediately process student enrollment for free courses without payment steps.
REQ-73	Checkout Initiator	The system shall initialize checkout transactions for paid courses, preventing duplicate active enrollments.
REQ-74	Duplicate Enrollment Check	The system shall verify that a student is not already enrolled before initiating checkout.
REQ-75	Enrolled Dashboard	The system shall display all actively enrolled courses on the student's dashboard.
REQ-76	Grant Content Player Access	The system shall authorize Course Player media streaming only for verified enrolled students.
REQ-77	Manual Admin Enrollment	The system shall allow Admins to manually grant course enrollment to a student account.
REQ-78	Cancel Enrollment	The system shall update enrollment status to Cancelled or Refunded upon refund confirmation.

Module 11: Stripe Payment Integration & Financial Ledger

ID	Requirement	Description
REQ-79	Stripe Hosted Checkout	The system shall construct secure Stripe Checkout sessions for course purchase operations.
REQ-80	Stripe Webhook Processing	The system shall handle asynchronous Stripe webhooks (checkout.session.completed) to verify transactions.
REQ-81	Create Payment Record	The system shall persist structured payment entities recording transaction ID, amount, and status.
REQ-82	Commission Calculation	The system shall split payment revenues into Platform Commission (20%) and Instructor Earnings (80%).
REQ-83	Payment Failure Handling	The system shall mark payments as Failed and notify students if transactions are declined.
REQ-84	Generate Invoice Receipt	The system shall generate downloadable PDF transaction invoices for completed purchases.
REQ-85	Currency Formatting	The system shall format pricing amounts in USD ($) while supporting localized currency conversions.
REQ-86	Financial Audit Log	The system shall record immutable log records for all financial events and status transitions.

Module 12: Refund Management Workflow

ID	Requirement	Description
REQ-87	Submit Refund Request	The system shall allow Students to request refunds within 14 days of purchase if course completion is under 20%.
REQ-88	Refund Queue	The system shall provide Admins with an inspection list of pending refund requests and progress metrics.
REQ-89	Approve Refund	The system shall allow Admins to approve requests, triggering Stripe refund APIs automatically.
REQ-90	Reject Refund	The system shall allow Admins to decline refund requests with explanatory reason text sent to the student.
REQ-91	Revoke Enrollment Access	The system shall automatically revoke student course player access upon refund approval.
REQ-92	Adjust Instructor Ledger	The system shall deduct refunded net earnings from the corresponding instructor balance ledger.

Module 13: Instructor Financials & Payout Operations

ID	Requirement	Description
REQ-93	Financial Analytics Dashboard	The system shall display gross earnings, net revenues, platform fees, and available balances for instructors.
REQ-94	Itemized Sales Ledger	The system shall render paginated sales logs showing date, student, course, and net earnings per sale.
REQ-95	Request Payout	The system shall allow Instructors to request earnings transfers when cleared balances exceed $50.00.
REQ-96	Process Payout	The system shall allow Admins to review and execute approved payout transfers to instructor payout accounts.
REQ-97	Payout Threshold Rule	The system shall enforce the minimum $50.00 balance constraint before enabling payout requests.
REQ-98	Export Financial CSV	The system shall allow Instructors and Admins to export sales and payout ledgers to CSV format.

Module 14: Learning Progress Tracking

ID	Requirement	Description
REQ-99	Auto-Create Lesson Progress	The system shall create a LessonProgress record when a student opens a lesson for the first time.
REQ-100	Mark Lesson Complete	The system shall mark lessons complete (CompletedAt=Timestamp) when video viewing threshold (90%) is passed.
REQ-101	Compute Course Progress %	The system shall compute overall course completion percentage based on completed vs. total lessons.
REQ-102	Save Playback Timecode	The system shall record periodic video timecode positions to resume playback where left off.
REQ-103	Progress Visual Bar	The system shall render visual progress percentage bars on student course cards and player UI.
REQ-104	Section Completion Track	The system shall mark sections complete once all underlying lessons are marked complete.

Module 15: Section Quizzes & Assessment

ID	Requirement	Description
REQ-105	Create Section Quiz	The system shall allow Instructors to attach an interactive quiz to any course section.
REQ-106	Configure Quiz Parameters	The system shall allow setting passing scores (%), time limits, and max attempt allowances.
REQ-107	Default Max Attempts	The system shall enforce a default limit of 3 allowed attempt attempts for section quizzes.
REQ-108	Automated Quiz Grading	The system shall automatically evaluate submitted answers against correct answer keys upon submission.
REQ-109	Display Instant Results	The system shall display score summaries, pass/fail status, and detailed feedback to students immediately.
REQ-110	Enforce Attempt Limits	The system shall block students from starting new attempts after exhausting allowed quiz attempts.

Module 16: Mandatory Final Examinations

ID	Requirement	Description
REQ-111	Create Mandatory Final Exam	The system shall allow Instructors to attach a single mandatory final examination to a course.
REQ-112	Single Attempt Constraint	The system shall strictly restrict mandatory final examinations to exactly one (1) attempt per student.
REQ-113	Unlock Examination Dependency	The system shall lock final examination access until 100% of course lessons are marked completed.
REQ-114	Final Exam Passing Rule	The system shall require passing the final exam as a mandatory prerequisite for certificate eligibility.
REQ-115	Instructor Assessment Analytics	The system shall display class average scores, pass rates, and common error analytics to instructors.
REQ-116	Lock Certificate Failure	The system shall withhold certificate generation if a student fails the single-attempt final exam.

Module 17: Questions & Multiple-Choice Options

ID	Requirement	Description
REQ-117	Add Question	The system shall allow Instructors to add multiple-choice questions with custom point values to quizzes/exams.
REQ-118	Add Answer Choices	The system shall allow adding multiple option choices per question, marking exactly one as correct.
REQ-119	Edit Question & Options	The system shall allow Instructors to update question text, point weighting, and answer choices.
REQ-120	Delete Question	The system shall allow Instructors to delete questions and cascade delete associated choice options.
REQ-121	Single Correct Answer Rule	The system shall validate that exactly one choice is set to IsCorrect=True before saving a question.

Module 18: Student Quiz Attempts & Logs

ID	Requirement	Description
REQ-122	Record Attempt Metadata	The system shall log attempt start time, submission time, duration, percentage score, and outcome.
REQ-123	Record Answer Choices	The system shall persist student question-by-question choice selections in StudentAnswers.
REQ-124	View Attempt History	The system shall allow Students to view their past attempt history, dates, and scores.
REQ-125	Instructor Attempt Audit	The system shall allow Instructors to inspect itemized student attempt logs and answers.
REQ-126	Reset Attempts Admin	The system shall allow Admins to reset student attempt logs for administrative exceptions.

Module 19: Course Reviews & Ratings

ID	Requirement	Description
REQ-127	Submit Review	The system shall allow enrolled students with >20% course completion to post a star rating (1-5) and review text.
REQ-128	Update Review	The system shall allow students to edit their previously submitted review and rating.
REQ-129	Moderate Review	The system shall allow Admins to delete offensive or spam student reviews from the platform.
REQ-130	Auto-Recalculate Rating	The system shall dynamically re-compute course average star ratings whenever reviews are added, updated, or removed.

Module 20: Automated Certificate Generation

ID	Requirement	Description
REQ-131	Verify Certificate Eligibility	The system shall verify 100% lesson completion and passed final exam before generating certificates.
REQ-132	Issue PDF Certificate	The system shall generate a branded PDF completion certificate with student name, course title, and date.
REQ-133	Public Verification Portal	The system shall provide a public verification page validating certificates by unique code.

Module 21: Live Sessions Management

ID	Requirement	Description
REQ-134	Schedule Live Session	The system shall allow Instructors to schedule live virtual sessions with external meeting links (Zoom/Meet/Teams).
REQ-135	Join Live Session	The system shall provide enrolled students with direct launch links to join scheduled live sessions.

3. Non-Functional Requirements

This section details all quality attributes, performance targets, and security parameters for EduHub.

NFR ID	Quality Attribute	Requirement Name	Detailed Technical Description	Measurable Acceptance Criteria
NFR-01	Performance	API Response Latency	System REST API endpoints must respond rapidly under normal operating conditions.	95% of standard read API requests must respond within < 200 ms.
NFR-02	Performance	Database Query Execution	Complex database queries (catalog search, dashboard aggregations) must utilize indexes.	Database execution time must not exceed 100 ms for 99% of queries.
NFR-03	Security	Password Hashing	Account passwords must be stored using industry-standard salted hashing algorithms.	Passwords hashed using BCrypt (work factor 12) or PBKDF2 with HMAC-SHA256.
NFR-04	Security	Encryption in Transit	All communications between clients, application server APIs, and gateways must be encrypted.	Enforce HTTPS TLS 1.3 encryption across all public and internal service endpoints.
NFR-05	Usability	Responsive Interface	Web UI must adjust fluently across desktop, tablet, and mobile viewports.	UI layout passes WCAG 2.1 AA and scales down to 360px viewport width seamlessly.
NFR-06	Usability	User Interface Feedback	The platform must provide immediate visual feedback during user actions.	Loading spinners or progress bars must render within 100 ms of user action.
NFR-07	Reliability	Automated Backups	Database entity records and media asset stores must be automatically backed up.	Incremental database backups every 6 hours; full daily backups retained for 30 days.
NFR-08	Availability	System Service Uptime	Core web application and REST APIs must maintain continuous availability.	Guaranteed service uptime of 99.5% per month, excluding scheduled maintenance windows.
NFR-09	Scalability	Concurrent Users	Architecture must support concurrent student learning sessions and video streams.	System supports 1,000 concurrent active users without degradation exceeding 10% latency.
NFR-10	Maintainability	API Specifications	All backend REST API endpoints must be documented using standardized OpenAPI specs.	100% of exposed REST endpoints documented in Swagger / OpenAPI 3.0 specification.
NFR-11	Data Integrity	ACID Transactions	All financial payments, enrollments, and quiz attempts must execute in ACID transactions.	Zero ledger discrepancies; automatic DB rollback occurs on partial transaction failures.
NFR-12	Compatibility	Cross-Browser Support	Platform web client must function identically across all major modern web browsers.	Full functional compatibility verified on Chrome, Firefox, Safari, and Edge.
NFR-13	Privacy	PII Protection	Platform must protect user personally identifiable information according to privacy laws.	Unencrypted PII is never logged; users can request complete data account deletion.
NFR-14	Error Handling	Standardized API Errors	Backend APIs must handle exceptions gracefully without exposing code stack traces.	Standard error payload format returned: { status, errorCode, message, timestamp }.

4. Use Case Diagrams & Descriptions

4.1 System Actors

Actor	Description
Visitor	Unauthenticated user browsing catalog, previewing sample lessons, and registering.
Student	Authenticated user enrolled in courses, taking quizzes/exams, streaming video content, and earning certificates.
Instructor	Authenticated content creator managing course syllabi, quizzes, live sessions, and financial earnings.
Admin	Administrative user managing accounts, taxonomy, course approval queues, refunds, and global settings.
External Payment Gateway (Stripe)	External service processing card transactions and sending asynchronous webhooks.
External Meeting Provider (Zoom / Google Meet / MS Teams)	Third-party video conferencing service hosting live learning sessions.

4.2 Standard Detailed Use Cases (UC-01 to UC-65)

Every use case below follows the same seven-section structure: related requirement, initiating actor, actor's goal, pre-condition, post-condition, main flow, and alternative flow.

UC-01: Register New User Account	
Related Requirement	REQ-01, REQ-02, REQ-08. Satisfies operational specifications for account creation.
Initiating Actor	Visitor
Actor's Goal	Create a new user account as a Student or Instructor on EduHub.
Pre-condition	Visitor has navigated to the EduHub registration page and is not currently authenticated.
Post-condition	A new unverified User record is created in the database, password is hashed, and an email verification token is dispatched.
Main Flow	1. The Visitor selects "Sign Up" on the EduHub navigation bar. 2. The system renders the registration form requesting full name, email address, password, and account role (Student/Instructor). 3. The Visitor inputs required details and submits the form. 4. The system validates that the email address is unique and password meets security complexity rules. 5. The system hashes the password using BCrypt and inserts the new User record with EmailConfirmed=False. 6. The system generates a cryptographic verification token and sends a verification link to the user's email address. 7. The system displays a confirmation message instructing the Visitor to verify their email.
Alternative Flow	4a. Email Already Exists: If the submitted email matches an existing account, the system halts registration, displays "Email address already registered," and provides a login link. 4b. Weak Password: If the password fails complexity requirements, the system displays specific guidance and prompts for a stronger password.

UC-02: Log In to Platform	
Related Requirement	REQ-03, REQ-07, REQ-08, REQ-10. Satisfies operational specifications for user authentication.
Initiating Actor	Visitor / User
Actor's Goal	Authenticate user credentials and secure access tokens to log in to EduHub.
Pre-condition	User possesses a registered EduHub account.
Post-condition	The user is authenticated, a JWT access token and refresh token are issued, and session context is established.
Main Flow	1. The Visitor selects "Log In" on the application header. 2. The system renders the authentication form requesting email address and password. 3. The Visitor enters their email and password, then submits the form. 4. The system retrieves the user record matching the email address and verifies the password hash. 5. The system checks that the account is active (IsActive=True). 6. The system generates a JWT access token containing user claims alongside a secure refresh token. 7. The system returns tokens to the client and redirects the user to their role-specific dashboard.
Alternative Flow	4a. Invalid Credentials: If the email is not found or password verification fails, the system displays "Invalid email or password." 5a. Account Suspended: If IsActive=False, the system blocks login and displays "Account suspended. Please contact platform support."

UC-03: Reset Forgotten Password	
Related Requirement	REQ-04, REQ-05. Satisfies operational specifications for password recovery.
Initiating Actor	Visitor / User
Actor's Goal	Reset a forgotten password using a secure token received via email.
Pre-condition	User has access to the email account associated with their EduHub registration.
Post-condition	User's password hash is updated in the database and active user tokens are revoked.

Main Flow	1. The Visitor clicks "Forgot Password?" on the login page. 2. The system displays the password recovery request page. 3. The Visitor enters their registered email address and submits the request. 4. The system verifies email existence, generates a password reset token valid for 15 minutes, and dispatches a reset link via email. 5. The Visitor opens the email link and is directed to the password reset page. 6. The Visitor enters a new password and submits the form. 7. The system validates token freshness, updates the user password hash, invalidates the reset token, and displays success.
Alternative Flow	4a. Email Not Found: To prevent account enumeration, the system displays "If this email is registered, a password reset link has been sent." 7a. Expired Token: If the token has expired (>15 minutes), the system displays "Password reset link has expired. Please request a new link."

UC-04: Log Out of Platform	
Related Requirement	REQ-06. Satisfies operational specifications for user session termination.
Initiating Actor	Student / Instructor / Admin
Actor's Goal	Safely terminate the active authenticated session on EduHub.
Pre-condition	User is logged into the platform with an active JWT token.
Post-condition	User access token is blacklisted, client session tokens are destroyed, and user is redirected to the home page.
Main Flow	1. The user selects "Log Out" from the profile dropdown menu. 2. The client application sends a logout request containing the JWT bearer token to the server. 3. The system adds the token identifier to the server revocation cache to prevent token reuse. 4. The system clears stored refresh tokens in the database for this session. 5. The client application deletes stored tokens from local storage. 6. The system redirects the user to the public landing page with a confirmation toast.
Alternative Flow	3a. Network Disconnection: If the server logout call fails due to network failure, the client forcibly destroys local tokens and redirects to the landing page.

UC-05: Manage User System Roles	
Related Requirement	REQ-09, REQ-10. Satisfies operational specifications for administrative role management.
Initiating Actor	Admin
Actor's Goal	Assign, modify, or revoke user roles within the platform.
Pre-condition	Admin is authenticated and possesses active Admin privileges.
Post-condition	User role mappings are updated in the UserRoles table, altering permissions for subsequent API calls.
Main Flow	1. The Admin navigates to the User Management section in the Admin Panel. 2. The system renders a paginated table of registered users. 3. The Admin searches for a specific user and selects "Edit Roles". 4. The system displays current assigned roles (Student, Instructor, Admin). 5. The Admin modifies role selections and clicks "Save Role Changes". 6. The system updates records in UserRoles and logs the action in audit logs. 7. The system displays a confirmation message confirming the role update.
Alternative Flow	5a. Self-Demotion Restriction: If an Admin attempts to revoke their own Admin role when no other active Admin exists, the system blocks the action and displays "Cannot revoke sole Admin role."

UC-06: Apply for Instructor Account	
Related Requirement	REQ-11. Satisfies operational specifications for instructor role applications.
Initiating Actor	Student
Actor's Goal	Submit an application to upgrade account privileges from Student to Instructor status.
Pre-condition	Student is logged in and has an active account with completed profile details.

Post-condition	An InstructorApplication record is created with status Pending and queued for Admin review.
Main Flow	1. The Student clicks "Become an Instructor" in the platform navigation menu. 2. The system presents the instructor application form requesting teaching experience, expertise, bio, and portfolio links. 3. The Student fills out required fields and submits the form. 4. The system validates required inputs and checks that no pending application exists for this user. 5. The system creates an InstructorApplication record with Status='Pending'. 6. The system notifies the Admin queue and displays an application confirmation to the Student.
Alternative Flow	4a. Duplicate Pending Application: If the student already has an application under review, the system disables submission and displays "Your application is currently under review."

UC-07: Review Instructor Applications	
Related Requirement	REQ-12. Satisfies operational specifications for reviewing instructor applications.
Initiating Actor	Admin
Actor's Goal	Evaluate pending instructor applications and approve or reject candidates.
Pre-condition	Admin is logged in and pending instructor applications exist in the queue.
Post-condition	Application status is updated; if approved, the target user is assigned the 'Instructor' role.
Main Flow	1. The Admin accesses "Instructor Applications" in the Admin Panel. 2. The system renders a list of pending applications ordered by submission date. 3. The Admin selects an application to inspect candidate credentials and portfolio details. 4. The Admin evaluates qualifications and selects "Approve" or "Reject". 5. Upon selecting "Approve", the system sets status to Approved, adds the 'Instructor' role to the user account, and dispatches a confirmation email. 6. The system refreshes the pending queue UI.
Alternative Flow	4a. Application Rejection: If the Admin selects "Reject", the system prompts for a rejection reason, updates status to Rejected, records the reason, and emails an explanation to the applicant.

UC-08: Deactivate or Reactivate Account	
Related Requirement	REQ-13, REQ-14, REQ-126. Satisfies operational specifications for administrative account management.
Initiating Actor	Admin
Actor's Goal	Deactivate a user account for policy violations or restore access to a deactivated account.
Pre-condition	Admin is logged in and targeting an existing user account.
Post-condition	Target account IsActive flag is toggled and user tokens are revoked/restored accordingly.
Main Flow	1. The Admin views a user profile in the Admin User Management dashboard. 2. The Admin selects "Deactivate Account" (or "Reactivate Account" if currently suspended). 3. The system prompts the Admin to confirm the action and specify an administrative reason code. 4. The Admin confirms the operation. 5. The system toggles the IsActive boolean flag in Users and invalidates active JWT/refresh tokens for that user. 6. The system records the action in audit logs and updates user status in the UI.
Alternative Flow	4a. Cancel Operation: If the Admin cancels the dialog, the system aborts without changing database records.

UC-09: Manage Profile Information	
Related Requirement	REQ-15, REQ-16, REQ-17. Satisfies operational specifications for personal profile updates.
Initiating Actor	Student / Instructor
Actor's Goal	Update personal profile details, biography, social links, and profile avatar image.
Pre-condition	User is logged in to EduHub.
Post-condition	Profiles database record and avatar image references are updated.

Main Flow	1. The user navigates to "Account Settings" -> "Edit Profile". 2. The system loads current profile details (display name, headline, bio, links, avatar). 3. The user updates text fields and/or uploads a new profile image file. 4. The system validates text fields and verifies that the image is a valid JPEG/PNG under 2MB. 5. The system uploads the avatar image to cloud storage and updates the profile entity. 6. The system returns updated profile details and displays "Profile saved successfully."
Alternative Flow	4a. Invalid Avatar File: If the user uploads an invalid file type or image >2MB, the system rejects the file and displays "Image must be JPEG/PNG under 2MB."

UC-10: Change Account Password	
Related Requirement	REQ-18. Satisfies operational specifications for authenticated password changes.
Initiating Actor	Student / Instructor / Admin
Actor's Goal	Change current account password while authenticated.
Pre-condition	User is logged in with an active session.
Post-condition	User's password hash is re-generated and updated in the database.
Main Flow	1. The user navigates to "Security Settings" -> "Change Password". 2. The system renders fields for Current Password, New Password, and Confirm New Password. 3. The user inputs current password and new password, then submits. 4. The system verifies that the current password matches the stored password hash. 5. The system verifies that the new password meets security requirements and matches confirmation. 6. The system hashes the new password using BCrypt and updates the user database entity. 7. The system displays a confirmation notification.
Alternative Flow	4a. Incorrect Current Password: If current password check fails, the system blocks the update and displays "Current password entered is incorrect."

UC-11: View Public Instructor Profile	
Related Requirement	REQ-19. Satisfies operational specifications for instructor profile discovery.
Initiating Actor	Visitor / Student
Actor's Goal	View public instructor profile, bio, expertise, ratings, and course portfolio.
Pre-condition	Target instructor account exists and is publicly accessible.
Post-condition	Public profile data and instructor's published courses are rendered on screen.
Main Flow	1. The actor clicks on an instructor's name or avatar from a course card or directory. 2. The system retrieves the instructor's public profile metadata, avatar, headline, bio, and social links. 3. The system queries published courses authored by this instructor along with enrollment stats and aggregate ratings. 4. The system renders the public instructor profile page displaying bio, metrics, and course catalog cards.
Alternative Flow	2a. Instructor Not Found: If target ID does not exist or corresponds to a deactivated account, the system renders a 404 page.

UC-12: Configure Notification Settings	
Related Requirement	REQ-20. Satisfies operational specifications for user notification preferences.
Initiating Actor	Student / Instructor
Actor's Goal	Manage email and system notification preferences.
Pre-condition	User is logged in to EduHub.
Post-condition	Preference flags in UserSettings table are updated.

Main Flow	1. The user navigates to "Account Settings" -> "Notification Preferences". 2. The system renders toggle options for Email Notifications (Course Updates, New Releases, Promotional, Quiz Alerts). 3. The user toggles desired notification settings and clicks "Save Preferences". 4. The system updates corresponding boolean flags in UserSettings. 5. The system displays a confirmation toast message.
Alternative Flow	No alternative flow applicable.

UC-13: Select Category Preferences	
Related Requirement	REQ-21, REQ-22, REQ-24. Satisfies operational specifications for category preferences.
Initiating Actor	Student
Actor's Goal	Select preferred course learning categories to personalize recommendations.
Pre-condition	Student is logged in to EduHub.
Post-condition	Records in StudentCategoryPreferences table are created or updated.
Main Flow	1. The Student accesses "Category Preferences" during onboarding or from settings. 2. The system renders available platform course categories with selection checkboxes. 3. The Student selects one or more preferred categories (e.g., Web Development, Data Science) and submits. 4. The system clears previously existing category preferences for this student. 5. The system inserts selected Category IDs into StudentCategoryPreferences. 6. The system displays a confirmation and updates recommendation feeds.
Alternative Flow	3a. No Category Selected: If student submits with zero categories selected, the system displays "Please select at least one category preference."

UC-14: View Recommended Courses	
Related Requirement	REQ-23, REQ-25. Satisfies operational specifications for personalized recommendations.
Initiating Actor	Student
Actor's Goal	Discover personalized course recommendations based on selected category preferences.
Pre-condition	Student is logged in and has configured category preferences.
Post-condition	Personalized course recommendations are fetched and rendered on the student dashboard.
Main Flow	1. The Student navigates to their student home dashboard. 2. The system reads preferred category IDs from StudentCategoryPreferences for this student. 3. The system queries published courses matching preferred categories in which the student is not yet enrolled. 4. The system orders results by course popularity and average rating. 5. The system renders the "Recommended For You" course carousel on the dashboard.
Alternative Flow	2a. No Saved Preferences: If the student has zero saved category preferences, the system fallback renders top overall trending courses.

UC-15: Browse and Filter Courses	
Related Requirement	REQ-26, REQ-27, REQ-28, REQ-29, REQ-31, REQ-33. Satisfies operational specifications for catalog browsing and filtering.
Initiating Actor	Visitor / Student
Actor's Goal	Search, filter, and browse published courses across categories, prices, and ratings.
Pre-condition	User is on the public course catalog page.
Post-condition	Catalog view updates with filtered, paginated course cards matching filter parameters.

Main Flow	1. The actor opens the Course Catalog page. 2. The system loads and displays the default paginated grid of published courses alongside filter sidebar controls. 3. The actor applies filter criteria (e.g., Category: "Design", Subcategory: "UI/UX", Price: "Free", Min Rating: 4.0). 4. The actor selects a sort order (e.g., "Highest Rated"). 5. The system executes a paginated dynamic database query incorporating filter parameters. 6. The system returns matching course records and updates the display grid dynamically.
Alternative Flow	5a. No Matching Courses: If no courses match applied filters, the system displays "No courses found matching your criteria" with a "Reset Filters" button.

UC-16: Search Courses by Keyword	
Related Requirement	REQ-30. Satisfies operational specifications for keyword search.
Initiating Actor	Visitor / Student
Actor's Goal	Search for specific courses by entering text search keywords.
Pre-condition	User is on any page containing the platform search bar.
Post-condition	Search result page renders courses whose title, subtitle, or description match keywords.
Main Flow	1. The actor enters search terms into the top header search bar (e.g., "React Development") and presses Enter. 2. The system sanitizes search string inputs. 3. The system executes a full-text search query across Title, Subtitle, and Description columns of published courses. 4. The system renders search results ordered by text relevance score.
Alternative Flow	2a. Empty Search Input: If search bar is submitted empty, the system redirects to the main catalog without applying search filters.

UC-17: View Featured Courses	
Related Requirement	REQ-32. Satisfies operational specifications for featured course banners.
Initiating Actor	Visitor / Student
Actor's Goal	Discover platform-promoted featured courses on the home page.
Pre-condition	Published courses flagged as featured exist in the database.
Post-condition	Featured course hero banner and cards are rendered on screen.
Main Flow	1. The actor visits the EduHub landing homepage. 2. The system executes a query for courses where IsFeatured=True and Status='Published'. 3. The system retrieves course thumbnails, titles, instructor details, ratings, and prices. 4. The system renders featured courses prominently in the landing page hero section.
Alternative Flow	2a. No Featured Courses: If no courses are flagged as featured, the system automatically falls back to displaying top overall enrolled courses.

UC-18: View Course Overview Details	
Related Requirement	REQ-34, REQ-35, REQ-37, REQ-39, REQ-40, REQ-85. Satisfies operational specifications for course detail view.
Initiating Actor	Visitor / Student
Actor's Goal	Inspect comprehensive course landing page details, syllabus structure, outcomes, and enrollment requirements.
Pre-condition	Target course exists in the system database.
Post-condition	Complete course landing page details are fetched and displayed.

Main Flow	1. The actor clicks on a course card from catalog, search, or recommendation lists. 2. The system queries course header details (title, subtitle, description, rating, student count, price). 3. The system queries section titles and lesson list structure (syllabus) without revealing streaming video links. 4. The system queries instructor profile details, target audience, and course prerequisites. 5. The system renders the complete course landing page with an "Enroll Now" or "Buy Now" call to action.
Alternative Flow	2a. Course Draft Access Attempt: If a non-owner/non-admin attempts to view a course in 'Draft' status, the system blocks access and renders a 404 page.

UC-19: Preview Sample Video Lessons	
Related Requirement	REQ-36. Satisfies operational specifications for free preview video playback.
Initiating Actor	Visitor / Student
Actor's Goal	Stream designated free sample video lessons before purchasing or enrolling in a course.
Pre-condition	Target lesson has IsFreePreview=True flag set by instructor.
Post-condition	Video player modal opens and streams public preview lesson video.
Main Flow	1. The actor clicks on a lesson marked with a "Preview" badge on the course syllabus breakdown. 2. The system verifies that IsFreePreview=True for the requested lesson ID. 3. The system generates a temporary public media streaming token for the preview lesson. 4. The system opens a video player modal and begins streaming the sample lesson video.
Alternative Flow	2a. Lesson Not Free Preview: If actor attempts to directly invoke video API for a locked lesson without active enrollment, the system returns HTTP 403 Forbidden.

UC-20: Read Student Course Reviews	
Related Requirement	REQ-38. Satisfies operational specifications for viewing course ratings and reviews.
Initiating Actor	Visitor / Student
Actor's Goal	Read published student star ratings, review comments, and overall rating distributions.
Pre-condition	Target course landing page is open.
Post-condition	Course ratings summary histogram and paginated review list are rendered.
Main Flow	1. The actor scrolls to the "Reviews & Ratings" section on the course detail page. 2. The system calculates the breakdown distribution of 1-star through 5-star ratings for the course. 3. The system fetches a paginated list of approved student reviews joined with reviewer student display names and avatars. 4. The system renders the rating summary histogram and review cards list.
Alternative Flow	3a. No Reviews Exist: If no reviews have been submitted yet, the system displays "No reviews yet. Be the first student to review this course!"

UC-21: Create Course Draft	
Related Requirement	REQ-41, REQ-42, REQ-43, REQ-44, REQ-45. Satisfies operational specifications for initializing course drafts.
Initiating Actor	Instructor
Actor's Goal	Create and edit a new course draft including basic metadata, pricing, cover image, and category mapping.
Pre-condition	User is logged in as an approved Instructor.
Post-condition	A new Courses entity with Status='Draft' is saved in the database.
Main Flow	1. The Instructor clicks "Create New Course" on their instructor dashboard. 2. The system initializes a new draft course entity assigned to the Instructor's ID. 3. The Instructor fills out course metadata (title, subtitle, description, language). 4. The Instructor configures pricing (setting price or marking as Free). 5. The Instructor assigns parent Category and Subcategory IDs. 6. The Instructor uploads a course thumbnail image banner. 7. The system validates inputs and updates the course draft record in the database.

Alternative Flow	6a. Missing Required Fields: If mandatory metadata fields (e.g. title, category) are missing, the system highlights incomplete sections and prevents draft submission.

UC-22: Submit Course for Admin Approval	
Related Requirement	REQ-46, REQ-50, REQ-70. Satisfies operational specifications for course submission.
Initiating Actor	Instructor
Actor's Goal	Submit completed course draft to Admin review queue for publication.
Pre-condition	Course draft contains at least one section, at least one lesson, and required metadata.
Post-condition	Course status transitions from 'Draft' or 'Rejected' to 'PendingApproval'; rejection logs are cleared if resubmitted.
Main Flow	1. The Instructor opens their draft course management view and selects "Submit for Review". 2. The system validates content completeness rules (thumbnail, price, sections, lessons, video content). 3. The system updates Course.Status to PendingApproval and clears any existing RejectionReason log. 4. The system queues the course in the Admin approval list and sends a confirmation email to the Instructor. 5. The system displays a "Course submitted for approval" notification.
Alternative Flow	2a. Validation Failure: If course lacks sections or lessons, the system blocks submission and displays "Course must contain at least one section and lesson before submission."

UC-23: Delete Course Draft	
Related Requirement	REQ-47. Satisfies operational specifications for draft course deletion.
Initiating Actor	Instructor
Actor's Goal	Permanently delete an unpublished course draft.
Pre-condition	Course is owned by logged-in instructor and is in 'Draft' status with zero enrollments.
Post-condition	Course entity and associated child sections/lessons are deleted from database.
Main Flow	1. The Instructor clicks "Delete Draft" on a course card in their dashboard. 2. The system prompts with a warning: "Are you sure you want to permanently delete this course draft?" 3. The Instructor confirms deletion. 4. The system verifies Status='Draft' and EnrollmentCount==0. 5. The system cascade deletes the course and associated draft child entities. 6. The system removes the course card from the instructor dashboard UI.
Alternative Flow	4a. Active Enrollments Conflict: If course has active enrollments, the system blocks deletion and displays "Cannot delete course with active student enrollments."

UC-24: Archive Published Course	
Related Requirement	REQ-48. Satisfies operational specifications for course archiving.
Initiating Actor	Instructor / Admin
Actor's Goal	Archive a published course to prevent new enrollments while retaining access for existing enrolled students.
Pre-condition	Course status is 'Published'.
Post-condition	Course Status transitions to Archived, removing it from public catalog search while preserving active student player access.
Main Flow	1. The actor selects "Archive Course" from course management options. 2. The system displays a confirmation dialog explaining that new enrollments will be disabled while existing students keep access. 3. The actor confirms archiving. 4. The system updates Course.Status to Archived. 5. The system removes the course from public search indexes and updates status badge in UI.
Alternative Flow	No alternative flow applicable.

UC-25: View Instructor Dashboard Courses	
Related Requirement	REQ-49. Satisfies operational specifications for instructor course management views.
Initiating Actor	Instructor
Actor's Goal	View management dashboard list of all authored courses with status indicators and student counts.
Pre-condition	User is logged in as an Instructor.
Post-condition	Comprehensive list of instructor's courses is rendered on screen.
Main Flow	1. The Instructor navigates to the Instructor Dashboard. 2. The system queries all courses where InstructorId matches the logged-in user. 3. The system fetches enrollment counts, total revenue, and status badges (Draft, Pending, Published, Rejected, Archived) for each course. 4. The system renders course management cards with quick edit controls.
Alternative Flow	2a. No Authored Courses: If the instructor has not created any courses yet, the system renders a welcome banner with a "Create Your First Course" button.

UC-26: Manage Course Sections	
Related Requirement	REQ-51, REQ-52, REQ-53, REQ-54. Satisfies operational specifications for syllabus section management.
Initiating Actor	Instructor
Actor's Goal	Add, edit, reorder, or delete module sections within a course syllabus.
Pre-condition	Instructor is editing a course owned by them in Draft or Rejected status.
Post-condition	Sections table entity records are created, updated, reordered, or deleted.
Main Flow	1. The Instructor opens the Course Curriculum Builder. 2. To Add: Instructor clicks "Add Section", inputs title, and submits. System creates Section entity with auto-sequenced SortOrder. 3. To Edit: Instructor updates section title inline and clicks save. System updates Section record. 4. To Reorder: Instructor drags and drops sections into new order. System updates SortOrder values across all sections. 5. To Delete: Instructor clicks delete on empty section. System removes Section entity.
Alternative Flow	5a. Delete Non-Empty Section: If instructor attempts to delete a section containing lessons, system prompts: "Deleting section will also delete all contained lessons. Proceed?"

UC-27: Manage Section Lessons	
Related Requirement	REQ-55, REQ-56, REQ-57, REQ-58. Satisfies operational specifications for lesson content management.
Initiating Actor	Instructor
Actor's Goal	Create, reorder, upload video media to, and configure preview settings for section lessons.
Pre-condition	Course section exists and is being edited by course owner.
Post-condition	Lessons table entity records and associated media references are saved.
Main Flow	1. The Instructor expands a course section and selects "Add Lesson". 2. The Instructor enters lesson title and content type. 3. The Instructor uploads video media file or embeds video streaming URL. 4. The system processes video upload, computes duration, and updates Lesson.VideoUrl and Lesson.Duration. 5. The Instructor toggles IsFreePreview switch if desired. 6. The system updates lesson entity and renders updated syllabus tree.
Alternative Flow	4a. Media Upload Interrupted: If video upload fails midway, system alerts instructor and allows retrying upload without losing lesson title data.

UC-28: Manage Course Categories	
Related Requirement	REQ-59, REQ-60, REQ-61. Satisfies operational specifications for root category taxonomy.
Initiating Actor	Admin
Actor's Goal	Create, update, or delete top-level course categories.
Pre-condition	Admin is logged into the Admin Panel.
Post-condition	Categories database table records are created, updated, or removed.
Main Flow	1. The Admin accesses "Category Management" in the Admin Panel. 2. System lists all existing parent categories. 3. Admin clicks "Create Category", enters Name, Slug, and Icon identifier, then submits. 4. System validates name uniqueness and inserts record into Categories table. 5. Admin can click "Edit" to modify category fields or "Delete" to remove an unused category.
Alternative Flow	5a. Delete Category with Dependent Courses: If Admin attempts to delete a category that contains child subcategories or assigned courses, system blocks deletion and displays "Cannot delete category with dependent subcategories or courses."

UC-29: Manage Course Subcategories	
Related Requirement	REQ-62, REQ-63, REQ-64. Satisfies operational specifications for child subcategory taxonomy.
Initiating Actor	Admin
Actor's Goal	Create, update, and delete child subcategories under parent categories.
Pre-condition	Target parent Category exists in the database.
Post-condition	Subcategories database table records are managed.
Main Flow	1. Admin selects a parent Category and clicks "Add Subcategory". 2. System presents subcategory input form. 3. Admin enters Subcategory Name and Slug, then submits. 4. System verifies uniqueness under selected parent category and saves Subcategory entity. 5. Admin can edit existing subcategories or delete subcategories that have zero linked courses.
Alternative Flow	5a. Delete Subcategory with Linked Courses: System blocks deletion if active courses are linked to the subcategory and displays "Subcategory in use by courses."

UC-30: Review Submitted Course Queue	
Related Requirement	REQ-65, REQ-66, REQ-70. Satisfies operational specifications for administrative course inspection.
Initiating Actor	Admin
Actor's Goal	Inspect submitted courses awaiting publication approval in the admin review queue.
Pre-condition	Courses with status 'PendingApproval' exist in system.
Post-condition	Admin reviews full course details, syllabus, and lesson videos to prepare for approval decision.
Main Flow	1. Admin opens "Course Approvals Queue" in the Admin Panel. 2. System displays table of pending courses sorted by submission timestamp. 3. Admin selects a course to enter "Inspection Mode". 4. System renders course details, instructor info, price, and syllabus video playback controls for Admin review. 5. Admin evaluates course quality against platform content guidelines.
Alternative Flow	2a. Approval Queue Empty: If no courses are pending approval, system displays "No courses currently awaiting review."

UC-31: Approve Course Publication	
Related Requirement	REQ-67, REQ-71. Satisfies operational specifications for publishing approved courses.
Initiating Actor	Admin
Actor's Goal	Approve a inspected pending course for public publication.

Pre-condition	Admin is reviewing a course in 'PendingApproval' status.
Post-condition	Course status updates to 'Published', publication date is set, and course appears in public catalog.
Main Flow	1. Admin inspecting pending course clicks "Approve & Publish". 2. System displays confirmation modal. 3. Admin confirms approval decision. 4. System sets Course.Status='Published' and records ApprovedAt timestamp. 5. System triggers notification service sending automated approval email to course instructor. 6. System updates catalog indexes to include new published course.
Alternative Flow	No alternative flow applicable.

UC-32: Reject Submitted Course	
Related Requirement	REQ-68, REQ-71. Satisfies operational specifications for course rejection with feedback.
Initiating Actor	Admin
Actor's Goal	Reject a submitted course and provide specific corrective feedback to the instructor.
Pre-condition	Admin is inspecting a course in 'PendingApproval' status.
Post-condition	Course status updates to 'Rejected', feedback is recorded in RejectionReason, and instructor is alerted.
Main Flow	1. Admin inspecting pending course clicks "Reject Course". 2. System opens dialog requesting mandatory feedback explanation detailing necessary corrections. 3. Admin inputs detailed feedback text and submits. 4. System sets Course.Status='Rejected' and saves text in RejectionReason. 5. System sends automated email containing rejection feedback to instructor.
Alternative Flow	3a. Empty Rejection Feedback: System blocks rejection submission if feedback text area is left empty, displaying "Rejection reason feedback is required."

UC-33: Unpublish Active Course	
Related Requirement	REQ-69. Satisfies operational specifications for unpublishing active courses.
Initiating Actor	Admin
Actor's Goal	Unpublish a currently active course due to terms violation or copyright dispute.
Pre-condition	Target course status is 'Published'.
Post-condition	Course status changes to 'Unpublished', hiding it from search while retaining existing student enrollments.
Main Flow	1. Admin navigates to published course record in Admin Panel and selects "Unpublish Course". 2. System prompts Admin for administrative unpublish justification. 3. Admin confirms unpublish action. 4. System sets Course.Status='Unpublished' and sends alert email to course instructor. 5. System updates search index, hiding course from catalog views.
Alternative Flow	No alternative flow applicable.

UC-34: Enroll in Free Course	
Related Requirement	REQ-72, REQ-74. Satisfies operational specifications for free course enrollment.
Initiating Actor	Student
Actor's Goal	Enroll immediately in a free course without undergoing checkout payment steps.
Pre-condition	Student is logged in and target course has IsFree=True and Status='Published'.
Post-condition	Active Enrollment entity record is created with PricePaid=0.00 and access is granted instantly.

Main Flow	1. Student clicks "Enroll Now - Free" button on free course landing page. 2. System verifies student is authenticated and checks that student is not already enrolled. 3. System verifies Course.IsFree == True. 4. System creates Enrollment record (StudentId, CourseId, EnrolledAt=Now, Status='Active', PricePaid=0.00). 5. System displays success modal and redirects Student directly into Course Content Player.
Alternative Flow	2a. Already Enrolled: System detects existing active enrollment, suppresses enrollment creation, and displays "Go to Course Player" button instead.

UC-35: Purchase Paid Course via Stripe	
Related Requirement	REQ-73, REQ-74, REQ-79, REQ-80, REQ-81, REQ-82, REQ-83, REQ-85. Satisfies operational specifications for online paid course checkout via Stripe.
Initiating Actor	Student
Actor's Goal	Purchase full access to a paid course using credit/debit card via secure Stripe checkout.
Pre-condition	Student is logged in and course is published with Price > 0.00.
Post-condition	Payment is processed, financial ledger records are created with platform/instructor revenue splits, and active enrollment is granted.
Main Flow	1. Student clicks "Buy Course" button on paid course detail page. 2. System verifies student is not already enrolled and initializes pending Payment transaction record. 3. System calls Stripe API to construct a Hosted Checkout Session containing course line items and metadata. 4. System redirects Student to secure Stripe Hosted Checkout page. 5. Student inputs valid payment card details on Stripe checkout page and submits payment. 6. Stripe processes transaction and sends asynchronous checkout.session.completed webhook event to EduHub server. 7. System validates webhook cryptographic signature, updates Payment.Status='Success', computes Platform Fee (20%) vs Instructor Earnings (80%), creates Enrollment record with Status='Active', and sends email purchase receipt.
Alternative Flow	5a. Payment Card Declined: Stripe displays card decline error to student. System receives payment failure notification, sets Payment.Status='Failed', and preserves failure audit log without granting enrollment.

UC-36: View Enrolled Student Dashboard	
Related Requirement	REQ-75, REQ-101, REQ-102, REQ-103. Satisfies operational specifications for enrolled student course management.
Initiating Actor	Student
Actor's Goal	View dashboard list of enrolled courses, completion percentages, progress bars, and resume learning shortcuts.
Pre-condition	Student is logged in to EduHub.
Post-condition	Enrolled courses dashboard view renders active courses with updated completion stats.
Main Flow	1. Student selects "My Learning" from main user menu. 2. System queries active Enrollments for logged-in StudentId. 3. For each course, system computes percentage completion based on LessonProgress records and retrieves last accessed lesson metadata. 4. System renders enrolled course cards with visual progress bars and "Resume Learning" buttons.
Alternative Flow	2a. Zero Active Enrollments: System renders empty state UI with a "Browse Catalog" button.

UC-37: Stream Course Video Lessons	
Related Requirement	REQ-76, REQ-99, REQ-100, REQ-101, REQ-103, REQ-104. Satisfies operational specifications for video lesson playback and progress tracking.
Initiating Actor	Student
Actor's Goal	Play course video lessons in the online course player while automated progress tracking records viewing status.

Pre-condition	Student has active enrollment status for target course.
Post-condition	LessonProgress records update, video timecode is saved, and course completion percentage updates.
Main Flow	1. Student opens Course Player for an enrolled course. 2. System verifies active enrollment and loads lesson navigation tree and default video stream. 3. System checks for existing LessonProgress record; if none exists, creates record with FirstOpenedAt=Now. 4. Video player seeks to saved LastTimecode integer. 5. Student watches video; system periodically sends heartbeat updates saving current playback position. 6. When video playback exceeds 90% duration or student clicks "Mark as Complete", system sets CompletedAt=Now in LessonProgress. 7. System recalculates overall course progress percentage and unlocks next sequential lesson.
Alternative Flow	2a. Access Without Enrollment: System blocks video access, returns HTTP 403 Forbidden, and redirects to purchase page.

UC-38: Admin Manual Student Enrollment	
Related Requirement	REQ-77. Satisfies operational specifications for administrative enrollment overrides.
Initiating Actor	Admin
Actor's Goal	Manually grant student course enrollment for support override or offline payment resolution.
Pre-condition	Target student and course exist in system.
Post-condition	Enrollment record flagged EnrolledByAdmin=True is inserted.
Main Flow	1. Admin selects "Manual Enrollment" in Admin Panel. 2. Admin inputs Student Email and Course ID. 3. System verifies student and course existence and checks for active enrollment. 4. Admin inputs administrative note and confirms manual enrollment. 5. System inserts Enrollment record with Status='Active' and EnrolledByAdmin=True. 6. System sends welcome email notification to student.
Alternative Flow	3a. Student Already Enrolled: System alerts Admin that student is already actively enrolled in target course.

UC-39: Cancel Student Enrollment Access	
Related Requirement	REQ-78. Satisfies operational specifications for revoking enrollment access.
Initiating Actor	Admin / System
Actor's Goal	Cancel active student enrollment access following refund issuance or policy violation.
Pre-condition	Active enrollment record exists for student and course.
Post-condition	Enrollment.Status is updated to 'Cancelled' or 'Refunded', instantly blocking course player access.
Main Flow	1. System (during refund execution) or Admin selects "Cancel Enrollment". 2. System verifies active enrollment record. 3. System updates Enrollment.Status to 'Cancelled'. 4. System invalidates course player authorization cache for target student/course pair. 5. System sends notification email informing student of access revocation.
Alternative Flow	No alternative flow applicable.

UC-40: Download Payment Invoice Receipt	
Related Requirement	REQ-84. Satisfies operational specifications for generating purchase invoices.
Initiating Actor	Student
Actor's Goal	Download official PDF billing receipt for a completed course purchase transaction.
Pre-condition	Completed purchase Payment record exists for target student.
Post-condition	PDF billing receipt document is dynamically compiled and streamed for download.

Main Flow	1. Student navigates to "Purchase History" on their account dashboard. 2. System renders list of past transactions with dates, course titles, amounts, and receipt buttons. 3. Student clicks "Download Invoice" for a specific transaction. 4. System queries payment details, student profile, tax lines, and course metadata. 5. System compiles dynamic PDF document containing official transaction breakdown and receipt code. 6. System streams PDF file payload to browser for download.
Alternative Flow	No alternative flow applicable.

UC-41: Audit Payment Transactions	
Related Requirement	REQ-86. Satisfies operational specifications for financial transaction audit logs.
Initiating Actor	Admin
Actor's Goal	Inspect platform financial ledger and audit individual transaction records.
Pre-condition	Admin is logged into Admin Panel.
Post-condition	Tabular transaction audit log renders with filtering options.
Main Flow	1. Admin opens "Financial Transactions Audit" section. 2. System renders paginated audit table displaying Transaction ID, Date, Student Name, Course Title, Amount, Platform Commission, Instructor Net, and Status. 3. Admin applies date range or status filters (Success, Failed, Refunded). 4. System filters financial dataset and displays detailed audit logs.
Alternative Flow	No alternative flow applicable.

UC-42: Submit Refund Request	
Related Requirement	REQ-87. Satisfies operational specifications for student refund requests.
Initiating Actor	Student
Actor's Goal	Submit formal refund request for an eligible course purchase within platform policy rules.
Pre-condition	Student purchased course within past 14 days and completed < 20% of course content.
Post-condition	RefundRequest record is created with Status='Pending' and queued for Admin review.
Main Flow	1. Student opens Purchase History and selects "Request Refund" next to eligible course transaction. 2. System evaluates automated eligibility parameters: purchase age (<= 14 days) and course completion percentage (< 20%). 3. System displays refund request form asking for primary refund reason. 4. Student fills out reason text and submits refund request. 5. System creates RefundRequest entity (PaymentId, StudentId, Reason, Status='Pending'). 6. System displays refund request confirmation to student.
Alternative Flow	2a. Refund Ineligible (Window Expired): If purchase age > 14 days, system disables request and displays "Refund window expired (over 14 days since purchase)." 2b. Refund Ineligible (Progress Exceeded): If completion >= 20%, system disables request and displays "Ineligible for refund due to course progress exceeding 20%."

UC-43: Review Pending Refund Requests	
Related Requirement	REQ-88. Satisfies operational specifications for administrative refund inspection.
Initiating Actor	Admin
Actor's Goal	Inspect student refund requests alongside progress metrics to make approval decisions.
Pre-condition	Pending RefundRequest entities exist in queue.
Post-condition	Admin evaluates request details and candidate eligibility metrics.
Main Flow	1. Admin navigates to "Refund Requests Queue" in Admin Panel. 2. System lists pending refund requests displaying student name, course, purchase date, reason, and computed consumption percentage. 3. Admin selects a request to inspect detailed transaction log and student learning activity timeline.

Alternative Flow	Queue Empty: System displays "No pending refund requests awaiting review."

UC-44: Approve Refund Request	
Related Requirement	REQ-89, REQ-91, REQ-92. Satisfies operational specifications for executing approved refunds.
Initiating Actor	Admin
Actor's Goal	Approve student refund, executing Stripe API refund payout, revoking access, and adjusting ledgers.
Pre-condition	Admin is reviewing pending refund request.
Post-condition	Stripe refund processes, RefundRequest.Status='Approved', course enrollment access is revoked, and instructor balance is adjusted.
Main Flow	1. Admin clicks "Approve Refund" on refund inspection card. 2. System prompts for confirmation. 3. Admin confirms approval. 4. System executes Stripe Refund API call using stored transaction payment intent ID. 5. Stripe confirms successful refund execution. 6. System updates RefundRequest.Status='Approved' and Payment.Status='Refunded'. 7. System triggers UC-39 revoking student enrollment access (Enrollment.Status='Refunded'). 8. System inserts negative financial ledger entry deducting refunded revenue from instructor's balance. 9. System sends automated refund confirmation email to student.
Alternative Flow	4a. Stripe Gateway Refund Failure: If Stripe refund API fails, system alerts Admin, aborts transaction rollback, and sets flag for manual gateway investigation.

UC-45: Reject Refund Request	
Related Requirement	REQ-90. Satisfies operational specifications for declining refund requests.
Initiating Actor	Admin
Actor's Goal	Decline student refund request with detailed explanation sent to student.
Pre-condition	Admin is inspecting a pending refund request.
Post-condition	RefundRequest.Status updates to 'Rejected', student retains course enrollment, and explanation email is sent.
Main Flow	1. Admin clicks "Reject Refund". 2. System opens dialog asking for mandatory rejection explanation message. 3. Admin inputs explanatory reason text and submits. 4. System updates RefundRequest.Status='Rejected' and stores explanation text. 5. System sends notification email to student detailing why refund request was declined.
Alternative Flow	3a. Empty Explanation: System blocks rejection submission until explanation text is provided.

UC-46: View Instructor Financial Earnings	
Related Requirement	REQ-93, REQ-94, REQ-98. Satisfies operational specifications for instructor financial dashboards.
Initiating Actor	Instructor
Actor's Goal	View total sales revenue breakdown, net earnings balance, pending payouts, and itemized course sales reports.
Pre-condition	User is logged in as an Instructor.
Post-condition	Instructor financial earnings dashboard renders balance summaries and sales tables.
Main Flow	1. Instructor opens "Financial Performance" tab on Instructor Dashboard. 2. System aggregates records from InstructorEarnings table for logged-in instructor. 3. System displays summary metrics cards: Total Gross Sales, Platform Fees, Total Net Revenue, Cleared Available Balance, and Pending Balance. 4. System renders itemized, paginated table of course sales including dates, student display names, course titles, and net earnings per sale.
Alternative Flow	No alternative flow applicable.

UC-47: Request Earnings Payout	
Related Requirement	REQ-95, REQ-97. Satisfies operational specifications for instructor earnings payout requests.
Initiating Actor	Instructor
Actor's Goal	Submit request to transfer available earnings balance to registered payout account.
Pre-condition	Instructor cleared available balance meets or exceeds minimum payout threshold ($50.00).
Post-condition	PayoutRequest record created with status 'Pending' and available balance updated.
Main Flow	1. Instructor views financial dashboard and clicks "Request Payout". 2. System verifies that cleared AvailableBalance >= 50.00. 3. System displays payout request modal confirming transfer amount and target payout method. 4. Instructor confirms payout request. 5. System inserts PayoutRequest entity (InstructorId, Amount, RequestedAt=Now, Status='Pending'). 6. System updates dashboard displaying pending payout status indicator.
Alternative Flow	2a. Balance Below Threshold: If available balance < $50.00, system disables "Request Payout" button and displays message "Minimum threshold of $50.00 required to request payout."

UC-48: Process Instructor Payout Request	
Related Requirement	REQ-96, REQ-98. Satisfies operational specifications for administrative payout processing.
Initiating Actor	Admin
Actor's Goal	Review pending instructor payout requests, release funds, and log completed payout transactions.
Pre-condition	Pending PayoutRequest records exist in Admin Panel.
Post-condition	Payout funds are transferred, PayoutRequest.Status='Completed', and instructor balance is adjusted.
Main Flow	1. Admin opens "Instructor Payouts Queue" in Admin Panel. 2. System displays list of pending payout requests. 3. Admin inspects request details and selects "Process Payout". 4. System triggers payout transfer via Stripe Payouts API / connected account transfer. 5. Upon payout processor confirmation, system updates PayoutRequest.Status='Completed', sets ProcessedAt=Now, and updates instructor account balance ledger. 6. System dispatches payout confirmation email to instructor.
Alternative Flow	4a. Payout Transfer Failure: System alerts Admin if payment gateway transfer fails, marks request 'Failed', and keeps funds in instructor available balance.

UC-49: Create Section Quiz	
Related Requirement	REQ-105, REQ-106. Satisfies operational specifications for section quiz creation.
Initiating Actor	Instructor
Actor's Goal	Attach an interactive assessment quiz to a course section and configure passing rules and attempt limits.
Pre-condition	Instructor is editing owned course section in Draft state.
Post-condition	Quiz entity record linked to target SectionId is created in database.
Main Flow	1. Instructor opens syllabus section editor and clicks "Add Section Quiz". 2. System opens quiz configuration dialog. 3. Instructor inputs Quiz Title, Passing Percentage (e.g. 70%), Time Limit (in minutes), and Maximum Allowed Attempts (default: 3 attempts). 4. Instructor saves configuration. 5. System inserts Quiz entity linked to SectionId with IsFinalExam=False. 6. System renders quiz block in section syllabus editor ready for question authoring.
Alternative Flow	4a. Section Already Has Quiz: System prevents creating multiple quizzes per section, displaying "Section already contains a quiz."

UC-50: Take Section Quiz	
Related Requirement	REQ-107, REQ-108, REQ-109, REQ-110, REQ-122, REQ-123. Satisfies operational specifications for student section quiz completion.
Initiating Actor	Student
Actor's Goal	Complete interactive section quiz assessment and view immediate scoring feedback.
Pre-condition	Student is actively enrolled in course, section quiz exists, and student has remaining attempt allowance (< max attempts).
Post-condition	QuizAttempt and StudentAnswers records are created, score is auto-calculated, and instant results review renders.
Main Flow	1. Student navigates to quiz lesson in Course Player and clicks "Start Quiz Attempt". 2. System verifies student attempt count < Quiz.MaxAttempts (default 3). 3. System creates QuizAttempt record (StartedAt=Now, Status='InProgress'), starts countdown timer, and presents questions payload. 4. Student selects multiple-choice option answers for each question and clicks "Submit Quiz". 5. System logs student answer choices into StudentAnswers table. 6. System automatically compares submitted options against correct answer keys, calculates percentage score, and determines pass/fail outcome (Score >= PassingScore). 7. System updates QuizAttempt record (SubmittedAt=Now, Score, Passed), updates section completion progress, and renders instant results screen with question explanations.
Alternative Flow	2a. Max Attempts Reached: System detects student has already used all allowed attempts (e.g. 3 attempts), disables quiz start button, and displays "Maximum attempt limit reached (3/3)." 4a. Time Limit Expired: If countdown timer reaches zero, system automatically auto-submits current selected choices and grades quiz attempt up to that point.

UC-51: Create Mandatory Final Examination	
Related Requirement	REQ-111, REQ-112. Satisfies operational specifications for mandatory final exam creation.
Initiating Actor	Instructor
Actor's Goal	Create mandatory comprehensive final course examination required for completion and certification.
Pre-condition	Instructor is editing an owned course draft.
Post-condition	Quiz entity flagged IsFinalExam=True and MaxAttempts=1 linked to CourseId is saved.
Main Flow	1. Instructor opens Course Assessment settings and selects "Create Mandatory Final Examination". 2. System opens final exam builder interface. 3. Instructor inputs Exam Title, Passing Score (e.g. 75%), and Time Limit (minutes). 4. System strictly sets IsFinalExam=True and hardcodes MaxAttempts=1 per approved business rules. 5. Instructor adds final exam comprehensive questions and choices. 6. Instructor saves final examination. 7. System attaches Final Exam entity directly to CourseId.
Alternative Flow	4a. Duplicate Final Exam: System enforces single final exam per course constraint, displaying "Course already possesses a mandatory final examination."

UC-52: Take Mandatory Final Examination	
Related Requirement	REQ-113, REQ-114, REQ-116, REQ-122, REQ-123. Satisfies operational specifications for completing mandatory final examination.
Initiating Actor	Student
Actor's Goal	Complete mandatory single-attempt final exam after completing 100% of video lessons to qualify for course certificate.
Pre-condition	Student has achieved 100% video lesson completion in course and has not yet attempted the single-attempt final exam.
Post-condition	Single attempt record is evaluated and saved; if passed, student qualifies for course certificate issuance.

Main Flow	1. Student completes final course lesson and selects "Take Mandatory Final Examination". 2. System verifies course lesson progress == 100%. 3. System verifies student has 0 prior attempts for this final examination. 4. Student clicks "Begin Final Examination". 5. System creates QuizAttempt record (IsFinalExam=True, AttemptNumber=1), starts countdown timer, and presents final exam questions. 6. Student completes and submits final exam responses. 7. System grades attempt instantly against answer keys, calculates final percentage score, updates attempt record, and displays official score result.
Alternative Flow	2a. Lessons Incomplete: If student attempts to access final exam before finishing all video lessons, system locks exam and displays "You must complete 100% of course video lessons before unlocking the final exam." 3a. Single Attempt Already Used: If student has already taken the single-attempt final exam, system displays "Final examination limit reached (1/1 attempt used)."

UC-53: View Quiz Performance Reports	
Related Requirement	REQ-115, REQ-125. Satisfies operational specifications for instructor assessment analytics.
Initiating Actor	Instructor
Actor's Goal	Analyze aggregate class performance, average quiz/exam scores, pass/fail percentages, and individual attempt details.
Pre-condition	Instructor is logged in and published course has active student quiz/exam attempts.
Post-condition	Statistical performance charts and student attempt logs render on screen.
Main Flow	1. Instructor opens "Assessment Analytics" for a selected course. 2. System queries QuizAttempts table aggregated by quiz/exam ID. 3. System computes metrics: Total Attempts, Pass Rate %, Average Score %, and Question Difficulty Index. 4. System renders graphical bar charts showing score distributions alongside searchable student attempt log tables.
Alternative Flow	No alternative flow applicable.

UC-54: Manage Quiz Questions and Options	
Related Requirement	REQ-117, REQ-118, REQ-119, REQ-120, REQ-121. Satisfies operational specifications for question bank authoring.
Initiating Actor	Instructor
Actor's Goal	Add, edit, reorder, or delete multiple-choice questions and answer choices for a quiz or exam.
Pre-condition	Instructor is editing an authored Quiz/Exam in Draft state.
Post-condition	Questions and Options database tables are updated; exactly one correct choice per question constraint is enforced.
Main Flow	1. Instructor opens Question Editor for a selected Quiz/Exam. 2. Instructor clicks "Add Question", enters question prompt text and point value. 3. Instructor adds multiple answer option choices and flags exactly one choice as IsCorrect=True. 4. System validates that exactly one choice is marked correct per question. 5. System inserts Question and child Option records into database. 6. Instructor can edit existing questions or click "Delete" to remove a question and its options.
Alternative Flow	4a. Invalid Correct Choice Count: If instructor flags zero choices or multiple choices as correct, system blocks saving and displays "Each multiple-choice question must have exactly one correct answer."

UC-55: View Quiz Attempt History	
Related Requirement	REQ-124, REQ-126. Satisfies operational specifications for viewing attempt history.
Initiating Actor	Student
Actor's Goal	Review past quiz/exam attempt dates, scores achieved, and pass/fail statuses.
Pre-condition	Student is logged in and has previously attempted quizzes in course.

Post-condition	Tabular attempt history log renders for student review.
Main Flow	1. Student clicks "Attempt History" on a quiz lesson page. 2. System queries QuizAttempts matching StudentId and QuizId. 3. System renders history table displaying Attempt Number, Date/Time, Score %, Time Elapsed, and Status (Pass/Fail).
Alternative Flow	No prior attempts: System displays "No previous attempts recorded."

UC-56: Submit Course Review and Rating	
Related Requirement	REQ-127, REQ-128, REQ-130. Satisfies operational specifications for student reviews.
Initiating Actor	Student
Actor's Goal	Leave star rating (1-5 stars) and written feedback review for an enrolled course after completing > 20% of content.
Pre-condition	Student has active enrollment with computed completion progress > 20%.
Post-condition	Review record is created or updated, and course AverageRating is auto-recalculated.
Main Flow	1. Student clicks "Leave a Review" on course player or course page. 2. System checks student progress; verifies completion > 20%. 3. System opens reviewsubmission modal with 5-star selector and text comment box. 4. Student selects star rating, inputs written review text, and submits. 5. System saves Reviewentity (StudentId, CourseId, Rating, Comment, CreatedAt=Now). 6. System automatically triggers background job recalculating course AverageRating (AVG(Rating)) and updates Courses record. 7. System displays review submission confirmation.
Alternative Flow	2a. Progress Insufficient: If student progress <= 20%, system blocks review and displays "You must complete at least 20% of the course before leaving a review."

UC-57: Moderate Course Reviews	
Related Requirement	REQ-129, REQ-130. Satisfies operational specifications for review moderation.
Initiating Actor	Admin
Actor's Goal	Remove abusive, offensive, or spam student reviews from the platform.
Pre-condition	Admin is inspecting reviews in Admin Panel.
Post-condition	Target Review entity is deleted and course average rating is re-computed.
Main Flow	1. Admin opens Review Moderation dashboard. 2. System lists reported or flagged student reviews. 3. Admin inspects review text and selects "Delete Review". 4. System prompts for administrative moderation justification. 5. Admin confirms deletion. 6. System deletes Review entity and recalculates course AverageRating.
Alternative Flow	No alternative flow applicable.

UC-58: Generate Course Certificate	
Related Requirement	REQ-114, REQ-131, REQ-132. Satisfies operational specifications for automated certificate issuance.
Initiating Actor	Student / System
Actor's Goal	Auto-generate and download official PDF completion certificate upon satisfying course completion rules.
Pre-condition	Student completed 100% of course video lessons AND passed mandatory final examination.
Post-condition	Certificate entity with unique verification code is created and downloadable branded PDF is compiled.

Main Flow	1. System evaluates completion criteria upon final exam pass: 100% lesson progress AND FinalExam.Passed == True. 2. System verifies eligibility and generates unique cryptographic verification code (e.g. CERT-EDUHUB-2026-X8921). 3. System creates Certificate entity (StudentId, CourseId, CertificateCode, IssuedAt=Now). 4. Student clicks "Download Certificate" in dashboard or player. 5. System compiles dynamic PDF certificate document displaying student name, course title, completion date, instructor signature, and verification URL/QR code. 6. System streams PDF certificate file to browser for download.
Alternative Flow	1a. Ineligible for Certificate: If student has not passed final exam or has incomplete lessons, system blocks certificate download button and displays requirements checklist.

UC-59: Verify Certificate Authenticity	
Related Requirement	REQ-133. Satisfies operational specifications for public certificate verification.
Initiating Actor	Visitor / Student
Actor's Goal	Verify legitimacy and authenticity of an issued EduHub certificate using unique certificate code.
Pre-condition	User has navigated to public Certificate Verification portal (/verify-certificate).
Post-condition	Verification portal displays official certificate confirmation details or invalid code warning.
Main Flow	1. Actor enters certificate code (e.g. CERT-EDUHUB-2026-X8921) into verification input bar and submits. 2. System queries Certificates table matching CertificateCode. 3. System joins student name, course title, instructor name, and issue date. 4. System renders official Verification Badge page displaying "Authentic Certificate Issued by EduHub" alongside issue details.
Alternative Flow	2a. Invalid Certificate Code: System fails to locate matching code and renders warning page: "Invalid Certificate Code. No record found matching this identifier."

UC-60: Schedule Live Learning Session	
Related Requirement	REQ-25, REQ-134. Satisfies operational specifications for live session scheduling.
Initiating Actor	Instructor
Actor's Goal	Schedule external live web learning session (Zoom / Google Meet / MS Teams) for course students.
Pre-condition	Instructor owns an active published course.
Post-condition	LiveSession entity record created and email notifications sent to enrolled students.
Main Flow	1. Instructor opens "Live Sessions" manager for a course and selects "Schedule Live Session". 2. Instructor enters Session Title, Description, Scheduled Start Date/Time, Duration, Provider (Zoom/Meet/Teams), and External Meeting Join URL. 3. System validates meeting URL formatting and future timestamp. 4. System inserts LiveSession entity (CourseId, Title, StartTime, ExternalUrl, Provider). 5. System sends automated email alerts to all actively enrolled course students and creates calendar event entry.
Alternative Flow	3a. Past Timestamp Selection: System blocks scheduling if selected start time is in the past, displaying "Session start time must be in the future."

UC-61: Join Live Learning Session	
Related Requirement	REQ-135. Satisfies operational specifications for joining live learning sessions.
Initiating Actor	Student
Actor's Goal	Access and launch external live learning session from within the course player.
Pre-condition	Student is actively enrolled in course and scheduled live session exists.
Post-condition	Student is securely redirected to external meeting join URL (Zoom/Meet/Teams).

Main Flow	1. Student views Live Sessions schedule tab in Course Player. 2. System displays upcoming live session details and launch control. 3. Student clicks "Join Live Session" button. 4. System verifies student's active enrollment status for course. 5. System logs session join activity and redirects browser window directly to external meeting join URL.
Alternative Flow	4a. Non-Enrolled Access Attempt: System blocks launch link for non-enrolled users and prompts course enrollment.

UC-62: Manage System Configuration	
Related Requirement	REQ-10, REQ-20. Satisfies operational specifications for global platform settings.
Initiating Actor	Admin
Actor's Goal	Configure global system parameters, platform commission split percentages, and email provider settings.
Pre-condition	Admin is logged into Admin Panel.
Post-condition	Global system setting key-value pairs are updated in database configuration cache.
Main Flow	1. Admin opens "System Settings" in Admin Panel. 2. System displays configurable parameters (Platform Fee %, Minimum Payout Threshold, Maintenance Mode toggle, SMTP Email settings). 3. Admin modifies parameter values (e.g. setting Platform Commission to 20%) and clicks "Save Configuration". 4. System validates parameter constraints and updates key-value entries in database configuration table. 5. System refreshes server application runtime configuration cache and displays success notification.
Alternative Flow	No alternative flow applicable.

UC-63: View Platform Analytical Dashboard	
Related Requirement	REQ-41, REQ-86, REQ-93. Satisfies operational specifications for administrative platform metrics.
Initiating Actor	Admin
Actor's Goal	View high-level platform analytical dashboard covering total users, active courses, enrollment totals, and gross revenue.
Pre-condition	Admin is logged into Admin Panel.
Post-condition	High-level platform executive metrics dashboard renders.
Main Flow	1. Admin opens Admin Overview Dashboard. 2. System aggregates database statistics: Total Registered Students, Total Approved Instructors, Total Published Courses, Monthly Active Enrollments, and Total Platform Gross Revenue. 3. System renders executive analytical dashboard featuring metric cards and historical performance charts.
Alternative Flow	No alternative flow applicable.

UC-64: Send System Announcements	
Related Requirement	REQ-20, REQ-71. Satisfies operational specifications for platform broadcast emails.
Initiating Actor	Admin
Actor's Goal	Broadcast system-wide email announcements to registered user segments.
Pre-condition	Admin is logged into Admin Panel.
Post-condition	Broadcast email job is queued and dispatched via email background service.

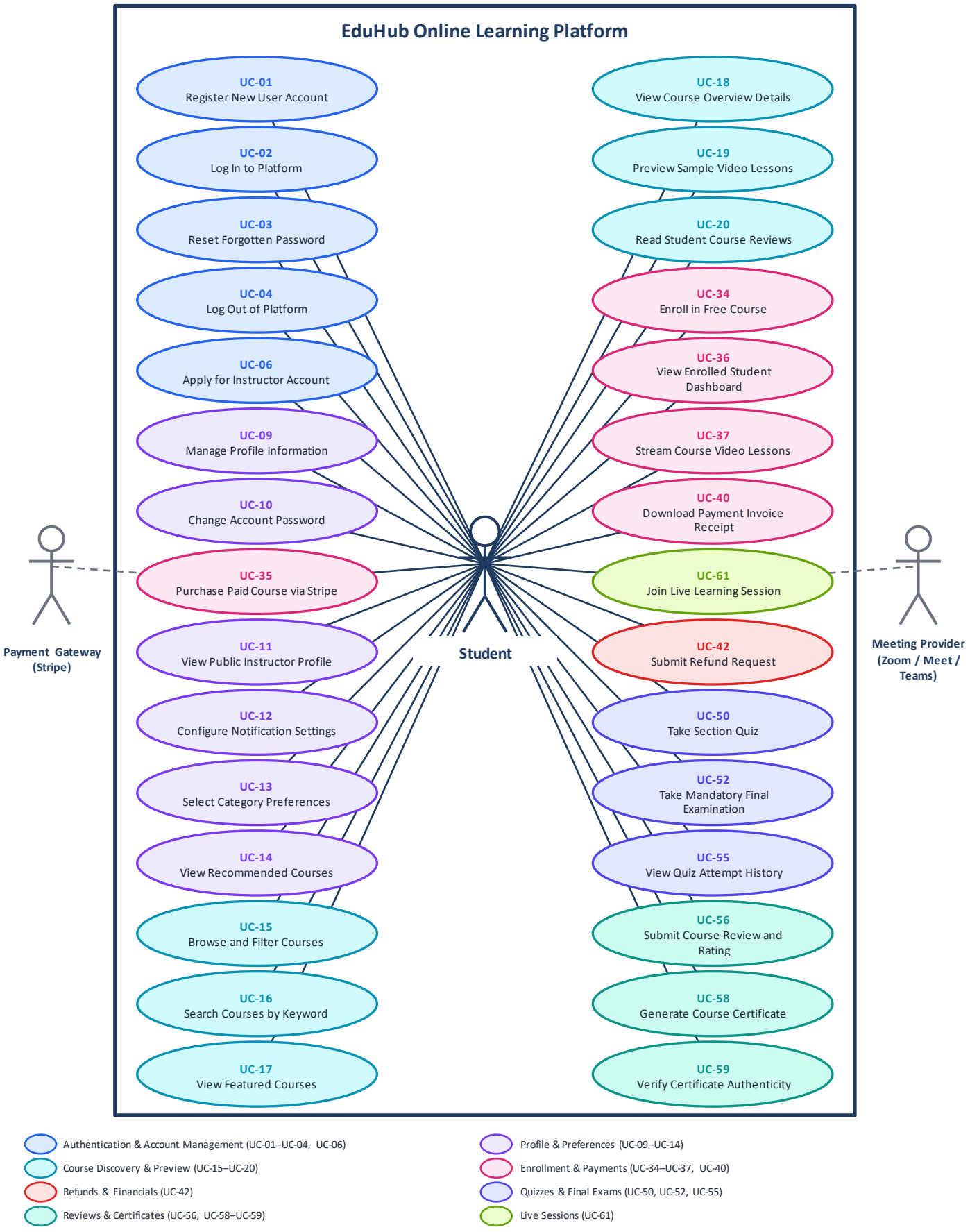
Main Flow	1. Admin selects "System Announcements" in Admin Panel. 2. Admin selects target user segment (All Users, All Students, All Instructors). 3. Admin inputs Email Subject and Rich-Text Email Content body. 4. Admin clicks "Send Announcement Broadcast". 5. System prompts for final broadcast confirmation. 6. Admin confirms; system queues background email jobs to dispatch emails to targeted user segment.
Alternative Flow	4a. Missing Subject or Body: System blocks broadcast execution if subject or content body is empty.

UC-65: Export System Audit Logs	
Related Requirement	REQ-10, REQ-86. Satisfies operational specifications for security audit log exports.
Initiating Actor	Admin
Actor's Goal	Export system security audit logs and administrative operation histories.
Pre-condition	Admin is logged into Admin Panel.
Post-condition	System audit log dataset is exported and downloaded as a formatted CSV file.
Main Flow	1. Admin opens "Security & Audit Logs" in Admin Panel. 2. System displays searchable audit event log table (logins, role changes, course status shifts, payouts). 3. Admin selects date filter range and clicks "Export Audit Logs CSV". 4. System queries audit log entries, formats records into CSV structure, and streams file payload for download.
Alternative Flow	No alternative flow applicable.

5. Use Case Diagrams

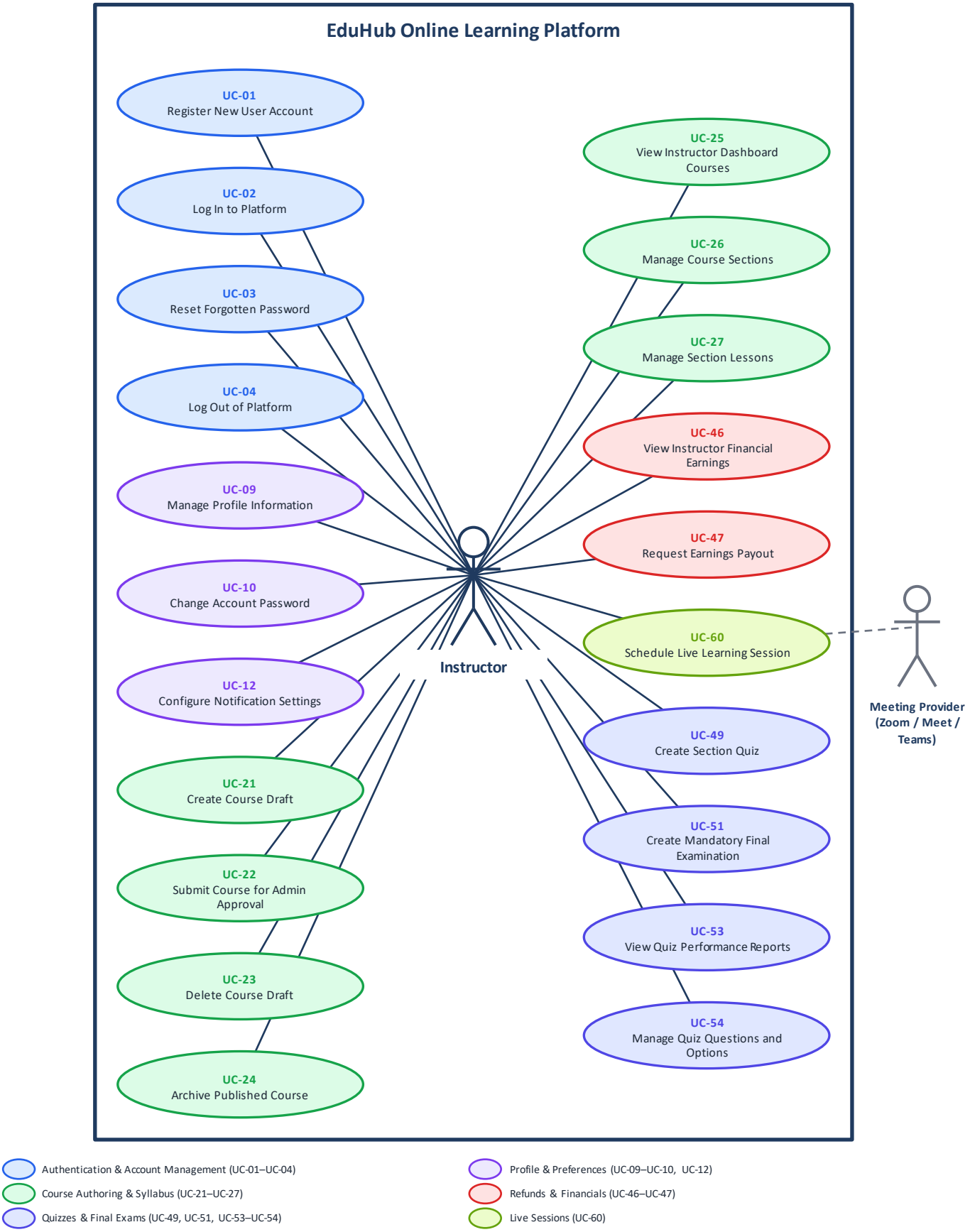
The following diagrams show the use cases initiated by each primary actor of the EduHub platform. Each use case is labeled with its identifier (UC-xx) and name as defined in Section 4.

5.1 Student Use Case Diagram



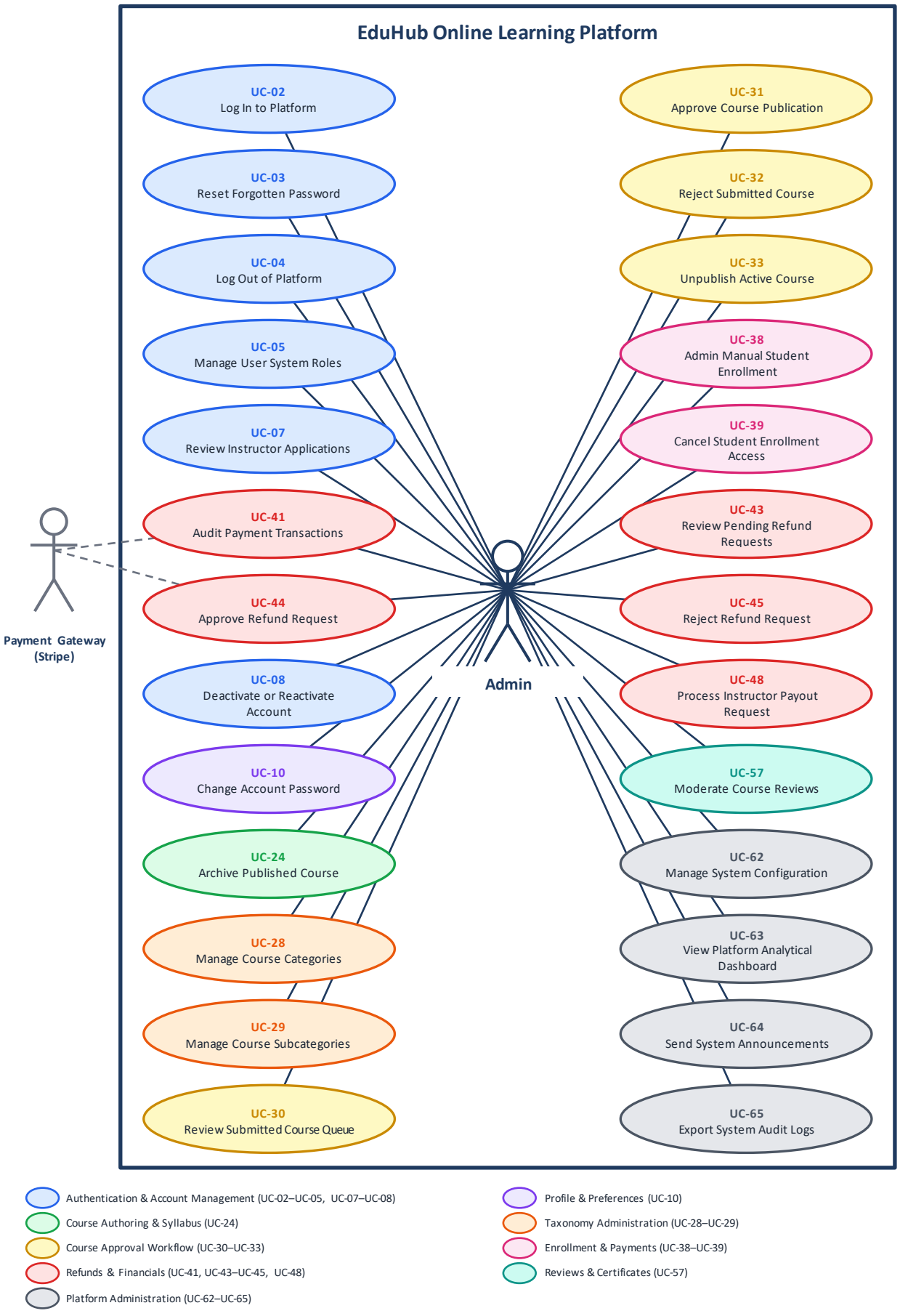
Total: 30 use cases shown for the Student diagram. Solid lines: association between the primary actor and a use case. Dashed lines: association with an external system actor.

5.2 Instructor Use Case Diagram



Total: 21 use cases shown for the Instructor diagram. Solid lines: association between the primary actor and a use case. Dashed lines: association with an external system actor.

5.3 Admin Use Case Diagram



Total: 26 use cases shown for the Admin diagram. Solid lines: association between the primary actor and a use case. Dashed lines: association with an external system actor.